Skills Summary

Intervals of Convergence with Endpoint Checks

Use this reference while completing your worksheet or when studying.

For $\sum a_n(x-c)^n$ the ratio test always returns an open interval $|x-c|<R$ and is *silent* at $|x-c|=R$. So the work is always the same four moves: ratio test for R, endpoints from $c \pm R$, substitute each endpoint to get a numeric series, then test that series with a tool the ratio test cannot supply.

Skill 1 — Ratio test and radius

Rule: form $|a_{n+1}/a_n|$, cancel, and factor out $|x-c|$. What remains is a number $L = \lim_{n\to\infty}$ of the x-free part. Convergence needs $|x-c| \cdot L < 1$, so $R = 1/L$ (and $R = \infty$ when $L = 0$). Divide top and bottom by the highest power of n to take the limit.

Example: Find R for $\displaystyle\sum_{n=1}^{\infty} \frac{(x-1)^n}{n \cdot 4^n}$.

Step 1. The terms carry a power of $(x-1)$, so the ratio test is the tool — build the ratio, cancel, and read off the x-free limit.

Step 2. $\left|\frac{a_{n+1}}{a_n}\right| = |x-1| \cdot \frac{n}{4(n+1)}$, and dividing by n gives $\frac{1}{4(1+1/n)} \to \frac{1}{4}$.

$\boldsymbol{L = \frac{1}{4}}$ so $R = 1/L = 4$, centered at $c = 1$.

Try it: Find L for $\displaystyle\sum_{n=1}^{\infty} \frac{(x+2)^n}{n \cdot 6^n}$.

check: $\boldsymbol{L = \frac{1}{6}}$

Skill 2 — Endpoints from center and radius

Rule: $|x-c|<R$ is the open interval from $c-R$ to $c+R$, so the two endpoints are exactly $c-R$ and $c+R$. Read c off the *written* form $(x-c)^n$: in $(x+2)^n$ the center is $c=-2$, not 2. An infinite radius has no endpoints at all.

Example: A series centered at $c = 1$ has radius $R = 4$. Find its endpoints.

Step 1. The ratio test gives the open set $|x-1| < 4$; its boundary is the pair of points where $|x-1|$ equals 4, so just add and subtract the radius.

Step 2. $c - R = 1 - 4$ and $c + R = 1 + 4$.

$\boldsymbol{x = -3}$ and $\boldsymbol{x = 5}$, so the open part is $-3 < x < 5$.

Try it: A series centered at $c = -2$ has radius $R = 6$. Give both endpoints.

check: $\boldsymbol{x = -8}$ and $\boldsymbol{x = 4}$

Skill 3 — Substituting an endpoint

Rule: put the endpoint value in for x and simplify until no x is left. The powers always collapse: $(x-c)^n$ becomes $(\pm R)^n$, which cancels the R^n in the denominator and leaves either 1 or $(-1)^n$ on top. That is the point of the substitution — it turns the power series into a numeric series you already know.

Example: Substitute the left endpoint $x = -3$ into $\sum_{n=1}^{\infty} \frac{(x-1)^n}{n \cdot 4^n}$ and find the exact sum.

Step 1. At an endpoint the ratio test is inconclusive, so substitute and identify the resulting numeric series by name.
Step 2. $x - 1 = -4$, so the terms are $\frac{(-4)^n}{n \cdot 4^n} = \frac{(-1)^n}{n}$ — the alternating harmonic series, convergent by the alternating series test.

$\mathbf{-\ln 2}$

Try it: Substitute $x = -3$ into $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n^2 \cdot 5^n}$ and give the exact sum.

check: $-\frac{\pi^2}{12}$

Skill 4 — Deciding at each endpoint

Rule: name the test. $\sum 1/n^p$ converges iff $p > 1$; an alternating series with terms decreasing to 0 converges; if the terms do not approach 0 the series diverges by the nth-term test. A converging endpoint earns a square bracket, a diverging one a round bracket — and the two endpoints are decided *separately*.

Example: Decide the right endpoint $x = 7$ of $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n^2 \cdot 5^n}$.

Step 1. Substitute, then match the numeric series to a test — here the sign pattern is constant, so a p-series comparison is the right tool.
Step 2. $x - 2 = 5$, so the terms are $\frac{5^n}{n^2 \cdot 5^n} = \frac{1}{n^2}$: a p-series with $p = 2 > 1$, so it converges, and $x = 7$ gets a square bracket.

$\boldsymbol{\frac{\pi^2}{6}}$

Try it: Decide the right endpoint $x = 4$ of $\sum_{n=1}^{\infty} \frac{(x-1)^n}{n \cdot 3^n}$: converges or diverges?

check: diverges

(!) Watch out: $a_n \to 0$ never proves convergence — the harmonic series is the standing counterexample. Test each endpoint separately: the two almost always behave differently, and a single bracket copied to both is the most common lost point on this topic.